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Mathematical model of the software development process according to the "outsource" scheme

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***Abstract* — The software development process is complex and multifaceted, influenced by a multitude of factors such as project complexity, collaboration among different teams, interaction with clients, and development methodologies. Efficiently managing this process becomes a crucial task for ensuring quality and meeting deadlines.**

In the context of the "outsourcing" scheme, where certain stages of software product development can be delegated to external specialists or teams, challenges arise in assessing the probabilistic-temporal characteristics of the development process. Questions arise about the expected time for each stage's completion, the likelihood of meeting set deadlines, and the factors exerting the greatest impact on process effectiveness.

The objective of this research is to develop a mathematical model that enables the analysis of probabilistic-temporal characteristics of software development processes within the "outsourcing" scheme. The model takes into account various factors, including participants' expertise level, communication effectiveness, and other parameters influencing stage durations.

The proposed mathematical model incorporates probabilistic transitions between stages, with transition probabilities and intensities depending on the involvement of external specialists and communication effectiveness. Relevant probabilistic distributions are used to estimate the duration of each stage, and statistical methods are employed to analyze probabilistic outcomes.

By considering the intricacies of the "outsourcing" scheme, this study aims to facilitate informed decision-making in resource allocation, project planning, and risk assessment within the domain of software development. Insights derived from the model's probabilistic-temporal characteristics provide valuable tools for enhancing process management and outcomes in the dynamic realm of software development.

Keywords* — *Software development, Outsourcing model, Probabilistic-temporal characteristics, Project complexity, GERT-model

I. Introduction

Computer systems and software have long become an integral part of the modern world, making a significant contribution to business processes, lifestyles, and technological advancement. The software development process is intricate and multifaceted. This process is influenced by numerous factors, such as project complexity, collaboration among diverse teams, interaction with clients, development methodologies, and more. Effective management of the software development process has become a pivotal task to ensure quality and meet deadlines.

In the context of the 'outsource' model, where a portion of software development can be outsourced to external specialists or teams, challenges arise in assessing the probabilistic-temporal characteristics of the development process. What is the expected time for different development stages? What is the likelihood of meeting prescribed deadlines? Which factors exert the greatest influence on the process's effectiveness?

The objective of this research is to develop a mathematical model that enables the analysis of probabilistic-temporal characteristics of the software development process within the 'outsource' model. The model should take into account the influence of various factors, including participants' expertise, levels of communication between stakeholders and developers, and other parameters impacting the duration of development stages.

II. State-of-the-Art

During the literature review, the following key aspects have been identified:

In [1], the results of a comprehensive overview of software development methodologies are presented, emphasizing the significance of the "outsource" model in contemporary contexts. This model offers more flexible resource management, providing access to experts worldwide. However, this work does not aim to analytically formalize software development processes and has a generalized nature.

Article [2] illustrates research findings highlighting the multitude of factors influencing the software development process in the "outsource" model. These factors encompass participants' professionalism, effective communication, project complexity, development methodologies, and other aspects. The importance of considering these factors is underscored. Nevertheless, the article does not directly discuss methodologies or approaches for incorporating these factors.

In [3], a mathematical model for outsourcing processes is implemented to optimize supply chain management. The overall supply chain cost is minimized by optimizing production and outsourcing quantities. The model employs a

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simple approach to describing time expenditures based on empirical knowledge, but such an approach has limitations. For instance, it cannot adequately describe non-sequential or complex processes. Additionally, it cannot assess probabilistic characteristics. Various approaches employing statistical methods, probability distributions, and analytical methods are needed.

In [4], authors emphasize the significance of assessing probabilistic-temporal characteristics, such as average time for stages completion, variance, and probability of meeting deadlines. These characteristics enable process management while considering the stochastic nature of variables. The modeling approaches proposed in the article are suitable for describing software development processes.

Article [5] focuses on the adaptability of models and management methods for software development in the face of dynamic external factors. Authors emphasize the importance of design and requirements documentation during development. These practices enable more flexible responses to changes and real-time decision-making. Unfortunately, the main source of information in this article is empirical knowledge, which reduces the practical value due to the absence of theoretical justification.

In [6], research is concentrated on resource allocation optimization methods in software development. This aids in finding optimal strategies under the "outsource" model for maximum efficiency. However, the authors do not evaluate probabilistic-temporal characteristics, preventing the consideration of numerous factors inherent in the software development process.

Considering the existing literature analysis, the purpose of this study is to develop a more precise and tailored mathematical model for analyzing probabilistic-temporal characteristics of the software development process within the "outsource" model, taking into account practical experience and factors influencing software development.

III. The software development team structure according to the "outsource" scheme

The Scrum team's composition is a critical element in Agile project management and plays a vital role in the successful delivery of the project. The team structure here delineates is derived from systematic analysis and follows widely accepted best practices. [7, 8] In this section, we elucidate the team members and their corresponding roles, along with the rational reasoning behind this particular structure.

A central tenet in Scrum is maintaining a team size of 7 +/- 2 members. This guideline is grounded in empirical studies showing that smaller teams foster better communication and collaboration, while larger teams may suffer from coordination overhead. Within this framework, our team comprises.

Business Analyst (BA): Tasked with understanding the business needs, translating them into actionable items, and ensuring that the development aligns with the strategic vision. This role bridges the gap between business and technical , thus minimizing misunderstandings and scope deviations.

Designer (D): Charged with crafting the user experience and visual aspects of the project. Incorporating a designer ensures that the product not only functions effectively but also appeals to the end-user, thereby enhancing market success.

Team / Tech Lead (TL) / Scrum Master: A vital position responsible for technical guidance, code quality assurance, and aligning the engineering team's efforts with the project goals. Their leadership guarantees a coherent development strategy. It is common that in addition a person is responsible for facilitating the Scrum process, resolving impediments, and ensuring that the team adheres to Scrum principles.

Back-End Engineer (BE): Engineers specializing in server-side operations, data management, and application logic. Their expertise ensures robust, scalable, and secure handling of the backend processes.

Two Front-End Engineers (FE): Dedicated to creating the interface with which the user interacts. Having two front-end engineers allows for a more efficient distribution of tasks and a collaborative environment that encourages innovation and quality.

Two Quality Assurance Engineers (QA): Responsible for verifying the quality of the product through systematic testing. Two QA engineers provide the necessary redundancy to ensure thorough examination and mitigate the risk of critical defects slipping into production.

The structural scheme of the software development team according to the "outsource" model is presented in Fig. 1.

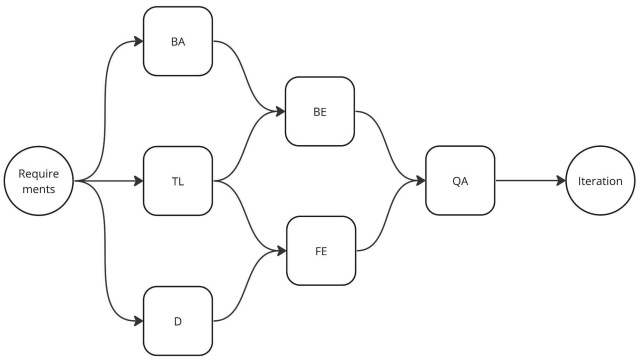

Fig. 1. The structural scheme of the software development team according to the "outsource" model.

The described composition is a manifestation of a well-balanced team that covers essential aspects of project development, from ideation to delivery. By adhering to the optimal team size, this structure minimizes communication complexities and fosters a collaborative environment.

IV. The features development flow

Due to the fact that Scrum is an approach in iterative software development we can create a model of Scrum iteration from Feature idea to Iteration which is the final result of software development flow.

The states during this flow are next one:

Feature Idea: This initial state represents the conception of a new functionality or enhancement. It's a nebulous stage where stakeholders envision a potential improvement or addition to the software.

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Business Logic / Requirements: Post conception, the feature idea undergoes a process of formalization. In this state, Business Analysts delineate the functional requirements, establishing a clear set of expectations and objectives for the proposed feature.

Technical Architecture: In this phase, Tech Lead formulate a technological roadmap. This entails deciding on design patterns, software components, interfaces, and other technical specifics crucial for the feature's realization.

Design / UI/UX: With a technical framework in place, the focus shifts to the user experience and interface. Designers sculpt the visual and interactive aspects, ensuring the software is intuitive, user-friendly, and aesthetically appealing.

Written Source Code: Developers translate the preceding stages into executable code, developing the functionality in accordance with the established requirements, architecture, and design.

Deployed on Server: Once code is written and undergoes preliminary testing, it's deployed to a server. This state signifies the transition of the feature from a development environment to a test or production environment, making it accessible for end-users or stakeholders.

Iteration: The culmination of all the preceding states, an iteration is a time-boxed period in which the developed features are integrated, tested, and reviewed. At the iteration's conclusion, stakeholders evaluate the delivered features, feeding back into the Scrum cycle for subsequent enhancements.

The transitions between those states have different weight based on amount of team members which are involved in the transition process.

This fact is clearly illustrated by the scheme of Figure 2.

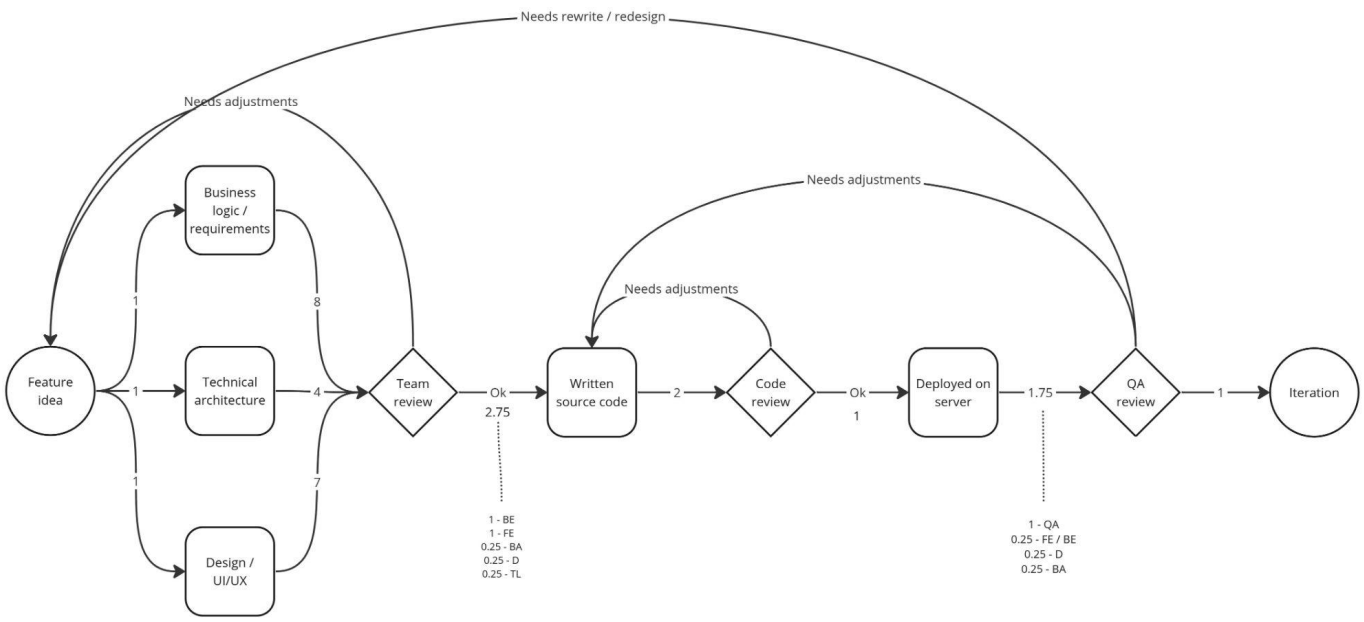

Fig. 2. Scheme of transitions between states in an "outsource" software development system.

Feature Idea to Business Logic / Requirements: The weight is 1, because only Business Analyst is involved.

Feature Idea to Technical Architecture: The weight is 1, because only Tech Lead is involved.

Feature Idea to Design / UI/UX: The weight is 1, because only the Designer is involved.

Business Logic / Requirements to Team review: The weight is 8, because the whole Team is involved in requirements review to understand what should be implemented. If some troubles will be identified during this transition then requirements need to be reworked.

Technical Architecture to Team review: The weight is 4, because only technical team members are involved in architecture discussion. If some troubles will be identified during this transition then architecture needs to be reworked.

Design / UI/UX to Team review: The weight is 7, because BE Engineer is not involved in the visual part of the system. If some troubles will be identified during this transition then design needs to be reworked.

Team review to Written Source Code: The weight is 2.75 due to the fact that BE & FE engineers are implementing the logic and BA, D & TL just helping them using their knowledge and clarifying needed points.

Written Source Code to Code review: The weight is 2 because reviewer and author of the code are both involved in the process of code analysis & possible adjustments. If some troubles will be identified during this transition then source code should be adjusted.

Code review to Deployed on Server: The weight is 1 because a single engineer is responsible for delivery the new logic to the server.

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Deployed on Server to QA review: The weight is 1.75 because a single QA is responsible for verifying the quality of the iteration and FE / BE, D & BA just helping them using their knowledge and clarifying needed points.

QA review to Iteration: The weight is 1 because a single person is responsible for delivery the iteration to the right server or other needed actions.

V. The software development process GERT model according to the "outsource" scheme

To uncover patterns in the software development process and determine probabilistic-temporal characteristics, a crucial task is the development of a mathematical process model. To address this task, we will employ the GERT modeling approach [9, 10]. The GERT diagram of the software development process according to the 'outsource' scheme is depicted in Figure 3.

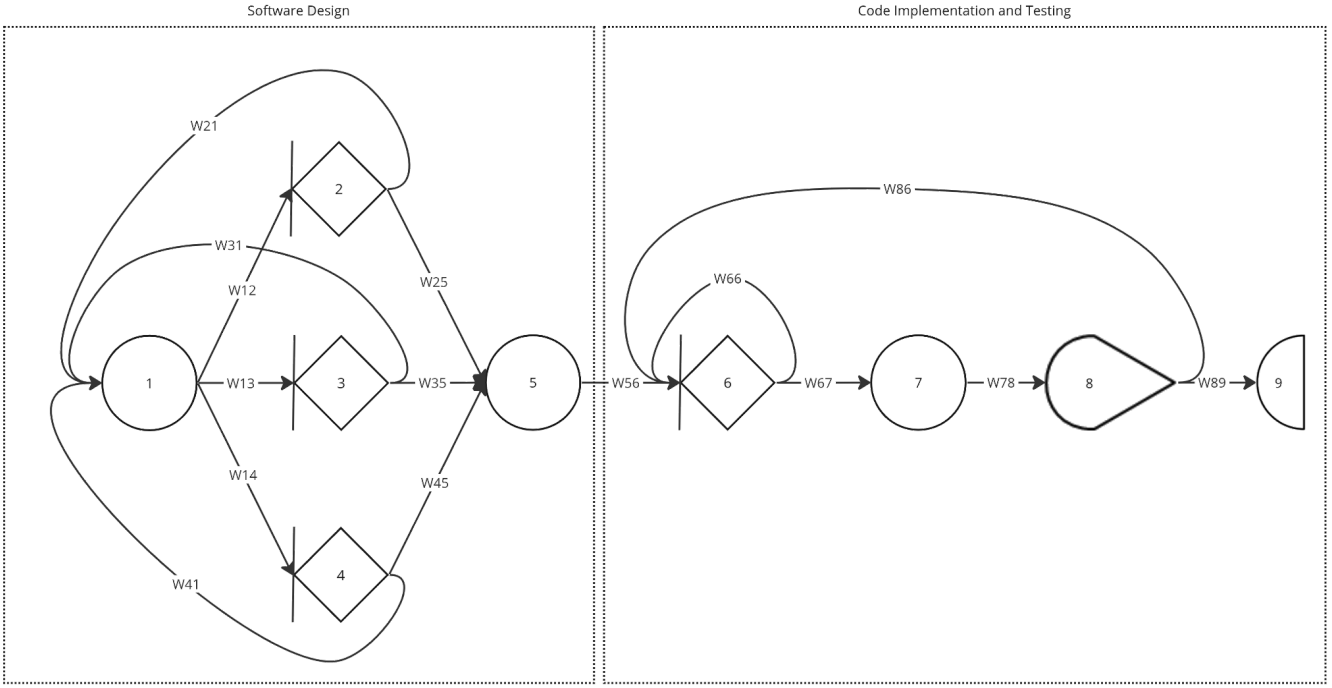

Fig. 3. The GERT diagram of the software development process according to the 'outsource'.

In accordance with the GERT diagram presented in Figure 3, the equivalent W-function of software development time can be represented as follows:

$$W_E(s) = \frac{\begin{pmatrix} W_{12}W_{25}W_{56}W_{67}W_{78}W_{89} + W_{13}W_{35}W_{56}W_{67}W_{78}W_{89} + \\ +W_{14}W_{45}W_{56}W_{67}W_{78}W_{89} \end{pmatrix}}{1 - \begin{pmatrix} W_{12}W_{21} + W_{13}W_{31} + W_{14}W_{41} + W_{12}W_{25}W_{56}W_{66} + \\ +W_{13}W_{35}W_{56}W_{66} + W_{14}W_{45}W_{56}W_{66} + \\ +W_{12}W_{25}W_{56}W_{67}W_{78}W_{86} + W_{13}W_{35}W_{56}W_{67}W_{78}W_{86} + \\ +W_{14}W_{45}W_{56}W_{67}W_{78}W_{86} \end{pmatrix}}$$

Due to the high analytical complexity of formalizing the presented model, we will decompose it and divide the diagram in Figure 3 into two components. We will refer to them as 1 - 'Software Design' and 2 - 'Code Implementation and Testing.'

It is worth noting that the interrelation between these two components will be formalized using an enhanced GERT model, where the application of fuzzy mathematics approaches would be appropriate.

Let's consider the first component (Software Design). At this stage, the formalization process of arcs 2-1, 3-1, and 4-1 essentially depends on a range of factors, most of which are related to the participants' level of professionalism. The higher the skill level of developers in terms of both software and hardware expertise, the better the communication among participants, including the project's client and stakeholders, the fewer the number of passes through the specified arcs, and therefore, the lower the costs for implementing this stage.

The described process is a culmination of actions involving communication and technical realization of developers' and clients' ideas and thoughts. This process is challenging to accurately describe with a single specific law of random distribution, as it depends on various factors, including project complexity, team dynamics, preferences of the parties involved, and more. However, an analogy can be drawn with a certain stochastic process, comparing it to the concept of a random walk.

In a random walk, an object moves in random directions with specific probabilities, and its final position depends on the sequence of random steps.

In the works of [11, 12], it has been noted that random walk processes can be described using a gamma distribution with variable coefficients k and θ:

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This assumption allows for describing random walk with considerations for the probabilistic characteristics of time intervals between steps.

It is well-known that the gamma distribution is a distribution of positive random variables, utilized for modeling time between events in stochastic processes or for modeling sums of random variables. It can exhibit a wide range of variability, spanning from exponential distribution to broader and asymmetric forms.

Transition characteristics between states in the 'Software Design' phase of the GERT network are presented in Table 1.

TABLE I. TRANSITION CHARACTERISTICS BETWEEN STATES IN THE "SOFTWARE DESIGN" PHASE OF THE GERT NETWORK

Branch	W-function	Transition Probability	Probability Density Coefficients
(1, 2)	W_{12}	P_1	$k=k_1$; $\theta=\theta_1$
(1, 3)	W_{13}	P_2	$k=k_1$; $\theta=\theta_1$
(1, 4)	W_{14}	$P_3=1-P_1-P_2$	$k=k_1$; $\theta=\theta_1$
(2, 5)	W_{25}	P_4	$k=k_2$; $\theta=\theta_2$
(3, 5)	W_{35}	P_5	$k=k_2$; $\theta=\theta_2$
(4, 5)	W_{45}	P_6	$k=k_2$; $\theta=\theta_2$
(2, 1)	W_{21}	$P_7=1-P_1$	$k=k_1$; $\theta=\theta_1$
(3, 1)	W_{31}	$P_8=1-P_2$	$k=k_1$; $\theta=\theta_1$
(4, 1)	W_{41}	$P_9=1-P_3$	$k=k_1$; $\theta=\theta_1$

Based on the provided data, the equivalent W-function of the "Software Design" phase can be represented as follows:

$$W_E(s)=\frac{W_{12}W_{25}+W_{13}W_{35}+W_{14}W_{45}}{1-\left(W_{12}W_{21}+W_{13}W_{31}+W_{14}W_{41}\right)}. \quad (1)$$

To "normalize" the expression, let's introduce the following variable:

$$p_{\bar{y}}^{\bar{d}}(t)=P_{d_1}\int_{-\infty}^{\infty}e^{isx}\frac{x^{k_{y_1}-1}\cdot e^{\frac{-x}{\theta_{y_1}}}}{\theta_{y_1}^{k_{y_1}}\cdot\Gamma\left(k_{y_1}\right)}\mathrm{d}x\times$$
$$\times P_{d_2}\int_{-\infty}^{\infty}e^{isx}\frac{x^{k_{y_2}-1}\cdot e^{\frac{-x}{\theta_{y_2}}}}{\theta_{y_2}^{k_{y_2}}\cdot\Gamma\left(k_{y_2}\right)}(x)\,\mathrm{d}x\times\ldots$$
$$\times P_{d_n}\int_{-\infty}^{\infty}e^{isx}\frac{x^{k_{y_n}-1}\cdot e^{\frac{-x}{\theta_{y_n}}}}{\theta_{y_n}^{k_{y_n}}\cdot\Gamma\left(k_{y_n}\right)}(x)\,\mathrm{d}x, \quad (2)$$

where $p_{\bar{y}}^{\bar{d}}(t)$ represents the product of the W-function; $\bar{d}=\left(d_1,d_2,d_3,..,d_n\right)$ – transition probability coefficients; $\bar{y}=\left(y_1,y_2,y_3,..,y_n\right)$ – coefficients of the generating function of transition moments.

The resulting expression for calculating the equivalent transfer functions of the "Software Design" phase can be formalized as:

$$W_E(t)=\frac{\left(p_{(1,2)}^{(1,4)}(t)+p_{(1,2)}^{(2,4)}(t)+p_{(1,2)}^{(3,4)}(t)\right)}{1-\left(p_{(1,1)}^{(1,7)}(t)+p_{(1,1)}^{(2,8)}(t)+p_{(1,1)}^{(3,9)}(t)\right)}. \quad (3)$$

In a similar manner, the resulting expression for calculating the equivalent transfer functions of the "Code Implementation and Testing" phase can be formalized.

Transition characteristics between states in the "Code Implementation and Testing" phase of the GERT network are presented in Table 2.

TABLE II. TRANSITION CHARACTERISTICS BETWEEN STATES IN THE "CODE IMPLEMENTATION AND TESTING" PHASE OF THE GERT NETWORK

Branch	W-function	Transition Probability	Probability Density Coefficients
(6, 7)	W_{12}	P_{10}	$k=k_1$; $\theta=\theta_1$
(7, 8)	W_{13}	P_{11}	$k=k_1$; $\theta=\theta_1$
(8, 9)	W_{14}	P_{11}	$k=k_1$; $\theta=\theta_1$
(6, 6)	W_{25}	$P_{12}=1-P_{10}$	$k=k_2$; $\theta=\theta_2$
(8, 6)	W_{35}	$P_{13}=1-P_{11}$	$k=k_2$; $\theta=\theta_2$

Based on the provided data, the equivalent W-function of the "Code Implementation and Testing" phase can be represented as follows:

$$W_E(s)=\frac{W_{67}W_{78}W_{89}}{1-\left(W_{66}+W_{67}W_{78}W_{86}\right)}. \quad (4)$$

The resulting expression for calculating the equivalent transfer functions of the "Code Implementation and Testing" phase can be formalized as:

$$W_E(t)=\frac{\left(p_{(1,1)}^{(1,11)}(t)\right)}{1-\left(p_{(2,2)}^{(10,12)}(t)+p_{(1,2)}^{(10,13)}(t)\right)}. \quad (5)$$

Using the probability density function formula [13, 14], we obtain the probability density distribution graph for the developed GERT network of the software development process in the case of P1=P2=0.3; P4=P5=P6=0.25; P10=P11=0.8. The results are shown in Figure 4.

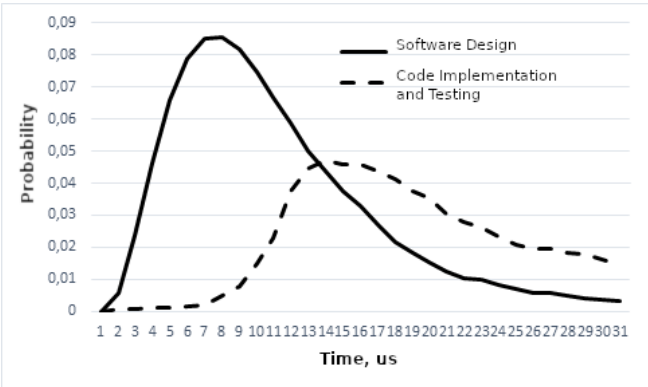

Fig. 4. Density distribution graphs of software development process execution time.

Mathematical expectations and variances of the obtained functions:

For Software design: Mean (μ) = 3.6 Variance (σ^2) = 129 for Code Implementation and Testing: Mean (μ) = 12.09 Variance (σ^2) = 231

VI. CONCLUSION AND FUTURE WORK

The paper presents a mathematical model of the software development process in accordance with the "outsourcing"

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scheme. The main scientific goal, which was to evaluate and investigate the probabilistic-temporal characteristics of software development under "outsourcing", is achieved.

The model is developed based on the GERT network modeling technique. It encompasses probabilistic transitions between states formed based on participants' competence levels and influencing factors. Notably, it enables the consideration of probabilistic aspects and dynamic changes over time.

The results of the analysis demonstrate that the proposed model provides valuable tools for evaluating probabilistic-temporal characteristics, resource management, and project planning. It helps to reduce risks and increase the efficiency of software development processes in "outsourcing" environment up to 5 % compared to the models [13, 14].

Thus, the developed mathematical model has potential for practical application in software development under "outsourcing" conditions. Its use can lead to more accurate process management, improved results and adaptation to dynamic changes in the modern development environment.

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